

DRY-RUN GUIDE | 09_bounded_stack.py
Source: Python\dsa_solutions\09_bounded_stack.py

CORE BEHAVIOR

Stack operations follow last-in-first-out order and enforce the configured capacity.

REPRESENTATIVE INPUT

Capacity 2; push 10, push 20, pop.

STEP TRACE

Stack [10,20]; pop removes top 20; stack is [10].

EXPECTED OUTPUT / RETURN

Popped 20; remaining [10].

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\dsa_solutions\09_bounded_stack.py; line numbers refer to the original file. No source was modified.

```
001 | import sys
002 |
003 |
004 | def main():
005 |     lines = sys.stdin.read().splitlines()
006 |     if not lines:
007 |         return
008 |     operation_count, capacity = map(int, lines[0].split())
009 |     stack = []
010 |     output = []
011 |     for line in lines[1:1 + operation_count]:
012 |         operation = line.split()
013 |         if operation[0] == "push":
014 |             if len(stack) == capacity:
015 |                 output.append("stack overflow")
016 |             else:
017 |                 stack.append(int(operation[1]))
018 |         elif operation[0] == "pop":
019 |             output.append(str(stack.pop()) if stack else "stack underflow")
020 |         elif operation[0] == "top":
021 |             output.append(str(stack[-1]) if stack else "stack underflow")
022 |     print("\n".join(output))
023 |
024 |
025 | if __name__ == "__main__":
026 |     main()
027 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.